

	Year Ended December 31,			2018 to 2019 % change	2017 to 2018 % change
	2019	2018	2017		
	(In thousands, except percentages)				
Cost of Revenue	\$ 234,282	\$ 181,255	\$ 127,422	29%	42%
Gross Margin	71.3%	69.7%	70.4%		

Cost of revenue increased \$53 million, or 29%, in 2019 compared to 2018. The overall increase was primarily due to increased employee compensation-related costs of \$23 million, driven by headcount growth, increased hosting costs of \$9 million, and increased third-party license fees of \$6 million. The increase in hosting costs was driven by an increase in expenditures for third-party managed hosting services, partially offset by lower depreciation from our self-managed colocation data centers in connection with our transition to third-party managed hosting services. The increase in third-party license fees was driven by increased customer usage of certain product features. Further contributing to the overall increase was an increase in allocated shared costs of \$4 million, primarily related to facilities and information technology.

Our gross margin increased by 1.6% in 2019 compared to 2018, driven primarily by increased optimization of our hosting infrastructure.

Cost of revenue increased \$54 million, or 42%, in 2018 compared to 2017. The overall increase was primarily due to increased hosting costs of \$21 million, increased employee compensation-related costs of \$20 million, and increased third-party license fees of \$7 million. The increase in hosting costs was driven by an increase in expenditures for third-party managed hosting services. The increase in employee compensation-related costs was driven by headcount growth. The increase in third-party license fees was driven by increased customer usage of certain product features. Further contributing to the overall increase was an increase in allocated shared costs of \$5 million, primarily related to facilities and information technology.

Our gross margin decreased by 0.7% in 2018 compared to 2017, driven primarily by increased hosting costs in connection with our transition to third-party managed hosting services while we continued to record depreciation for our existing self-managed colocation data centers. The overall decrease was partially offset by lower amortization of capitalized internal-use software due to timing of projects.

Operating Expenses

Research and Development Expenses

	Year Ended December 31,			2018 to 2019 % change	2017 to 2018 % change
	2019	2018	2017		
	(In thousands, except percentages)				
Research and Development	\$ 207,548	\$ 160,260	\$ 115,291	30%	39%

Research and development expenses increased \$47 million, or 30%, in 2019 compared to 2018. The overall increase was primarily due to increased employee compensation-related costs of \$34 million, driven by headcount growth, and increased allocated shared costs of \$8 million.

Research and development expenses increased \$45 million, or 39%, in 2018 compared to 2017. The overall increase was primarily due to increased employee compensation-related costs of \$36 million, driven by headcount growth, and increased allocated shared costs of \$6 million.

Sales and Marketing Expenses

	Year Ended December 31,			2018 to 2019 % change	2017 to 2018 % change
	2019	2018	2017		
	(In thousands, except percentages)				
Sales and Marketing	\$ 396,514	\$ 291,668	\$ 211,918	36%	38%